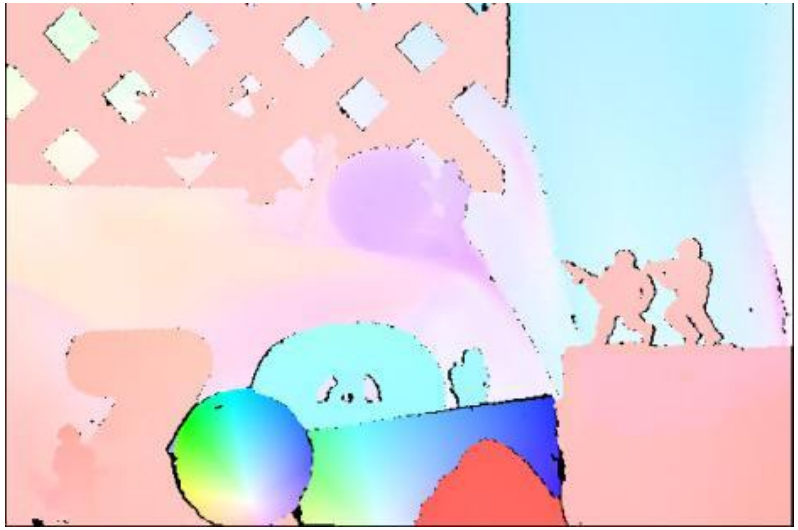
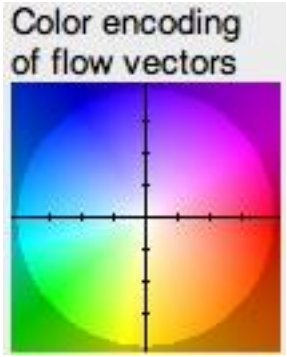
Flow quality evaluation

- Middlebury flow page
  - http://vision.middlebury.edu/flow/



Ground Truth



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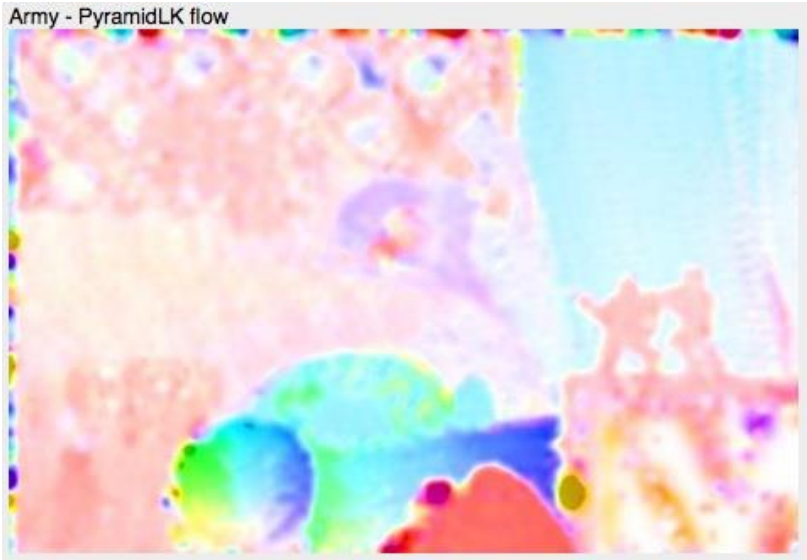
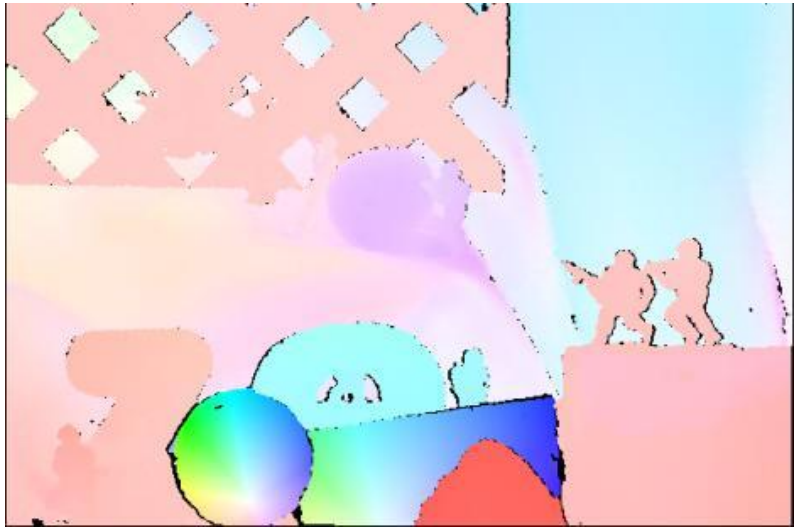
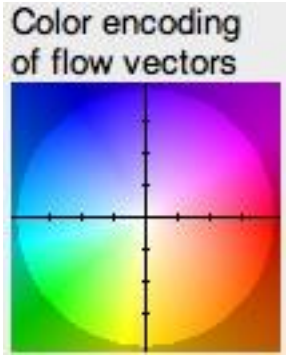


Lucas-Kanade flow



Ground Truth



Horn & Schunck algorithm

Additional smoothness constraint :

$$e_s = \iint ((u_x^2 + u_y^2) + (v_x^2 + v_y^2))dx\,dy,$$

besides the optical flow constraint equation term

$$e_c = \iint (I_x u + I_y v + I_t)^2 dx\,dy,$$

minimize $e_s + \alpha e_c$

CNN for Optical Flow Prediction

CNNs succeed in a variety CV tasks but not Optical Flow Prediction till the debut of ***FlowNet*** in 2015.

Dosovitskiy, Alexey, et al. "Flownet: Learning optical flow with convolutional networks." *Proceedings of the IEEE international conference on computer vision*. 2015.

- Learning to match not just learning feature representation
- Existing Optical Flow datasets too small for CNN training.
- Extreme high cost of ground truth annotation.
- ……

Traditional Optical-Flow Datasets

KITTI (194 training pairs)

- Large displacement; Special motion type
- **GT:** Recording the real-world scenes with camera & 3D laser scanner.

MPI Sintel (1041 training pairs)

- Small and large displacement.
- Two versions : Clean / With effects (blur / fog etc.)
- **GT:** Artificial scenes rendered with special attention to realistic image properties

Still too small for CNN training !

Synthetic Optical-Flow Datasets

Flying Chairs (22,872 pairs)

- Integrating 3D chair models into background images crawled from Flickr.
- **First** optical flow dataset designed for CNN training, proposed along with *FlowNet* in 2015.
- Surprisingly generalize well to traditional dataset like **KITTI**.

FlyingThings3D (more than 39,000 pairs)

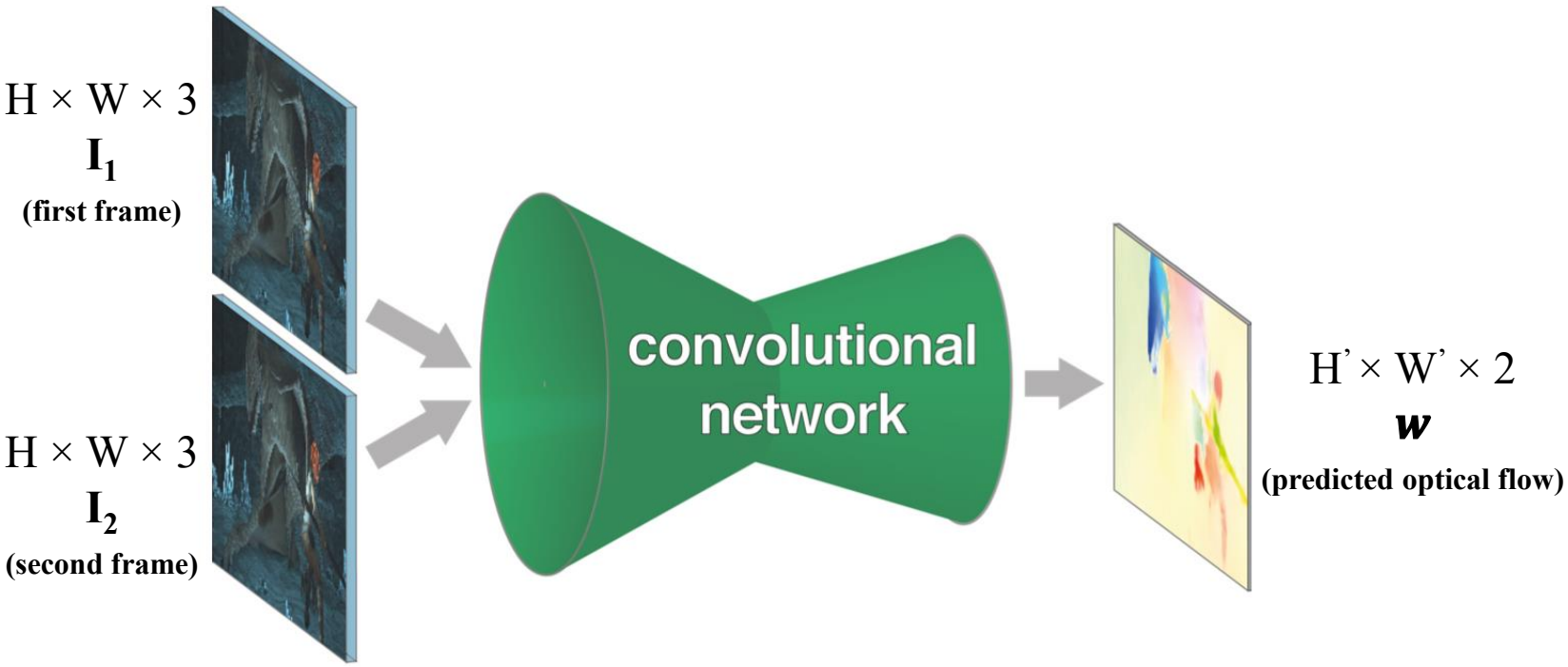
Mayer, Nikolaus, et al. "A large dataset to train convolutional networks for disparity, optical flow, and scene flow estimation." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.

- More than 39000 stereo frames in 960x540 pixel resolution, rendered from various synthetic sequences.

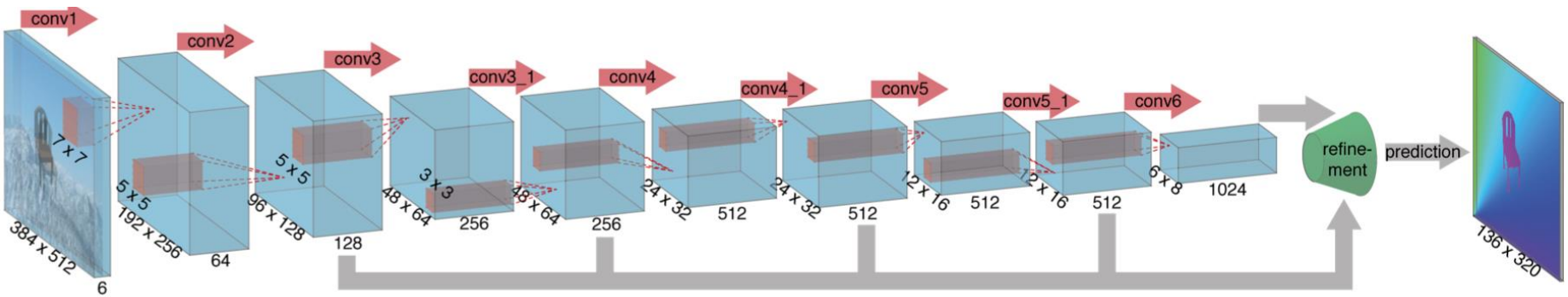
ALL integrated into Torchvision for easy use!

https://pytorch.org/vision/main/datasets.html#optical-flow

Overall Arch

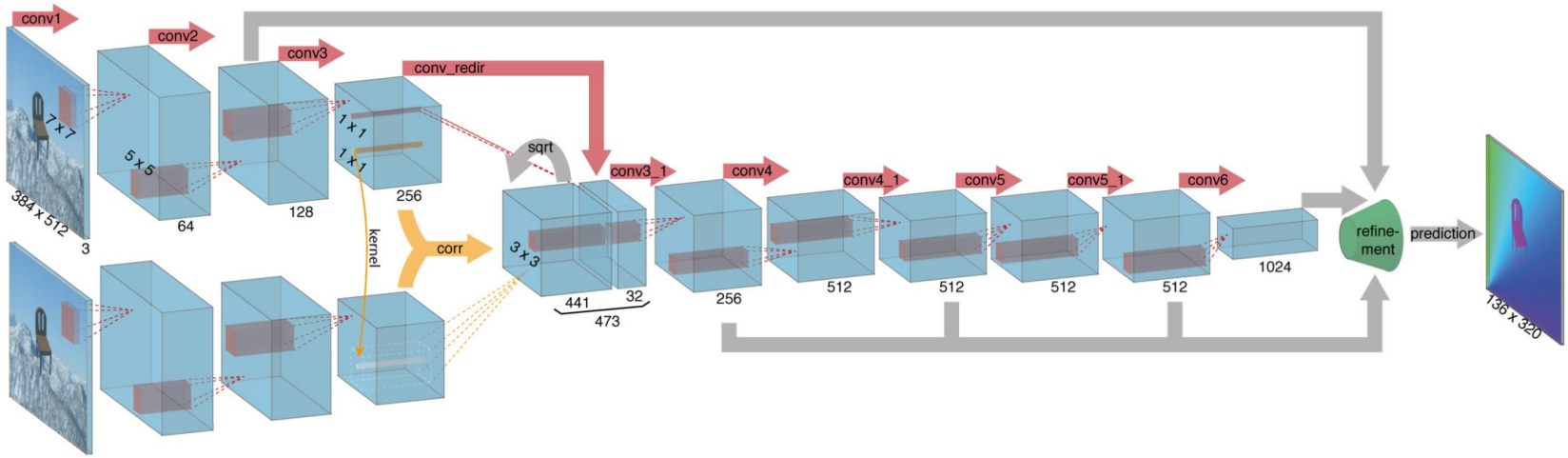


FlowNet-S (Simple)



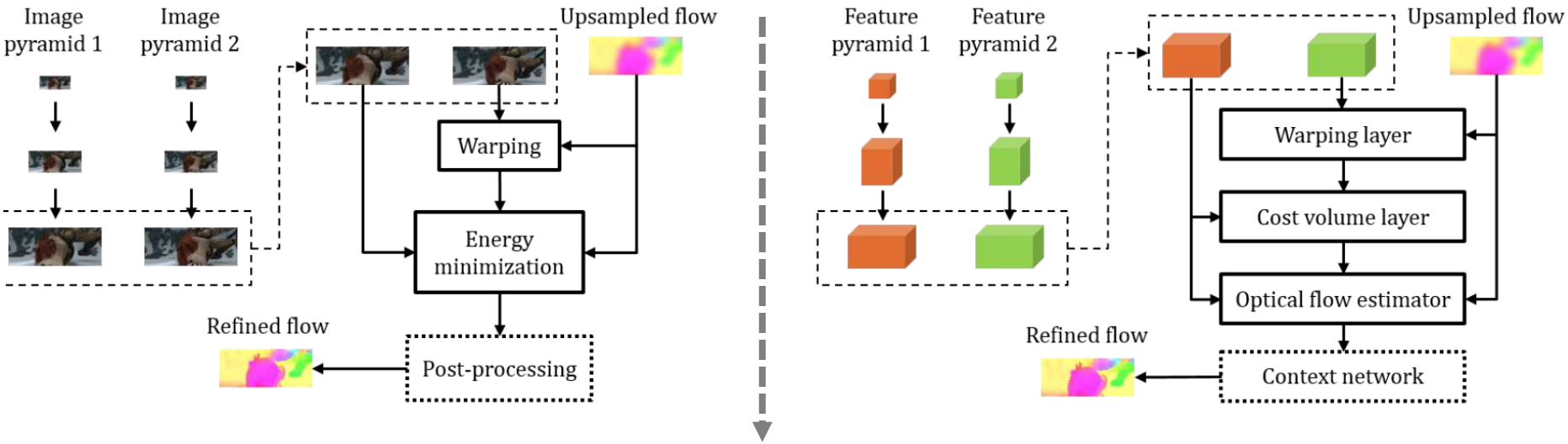
- Stack both input images (I_1 & I_2) to form the shape of $H \times W \times 6$.
- Pass through conv layers to extract feature maps at different level.
  (Features map size shrinks while depth increases)
- Feed certain levels of feature maps into the **refinement component**.
- The **refinement component** outputs the final Optical Flow

FlowNet-C (Correlation)



- Two separate yet identical processing streams for the two images.
- Combine them through correlation layer.
- Correlation layer explicitly provides matching capabilities.

PWC-Net



Traditional coarse-to-fine approach **PWC-Net**

Integrate 3 classic principles used in Traditional approach into CNN design:

1. Pyramid design
2. Wrapping
3. Postprocessing

Sun, Deqing, et al. "Pwc-net: Cnns for optical flow using pyramid, warping, and cost volume." Proceedings of the IEEE conference on computer vision and pattern recognition. 2018.